Bahadır Utku Kesgin

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Education

Koç University — Istanbul, Turkey
B.Sc., Electrical and Electronics Engineering — CGPA: 3.46/4.00
Track: Electronics and Waves — Track GPA: 4.00/4.00
Track: Quantum and Photonics — Track GPA: 3.85/4.00
Comprehensive National Athlete Scholarship (Turkish Ministry of Youth and Sports) — Expected Graduation: June 2027

Research Interests

Nonlinear and ultrafast photonics; optical and neuromorphic computing; spatiotemporal mode-locked fiber lasers; spatiotemporal dynamics in multimode fibers, guided wave nonlinear optics and optical chaos.

Research and Professional Experience

Koç University — Laboratory of Photonic Technologies (PI: Dr. Uğur Teğin) — Istanbul, Turkey
Undergraduate Researcher — Feb 2023 – Present

- Lead author on optical-computing architectures that exploit nonlinear and chaotic spatiotemporal dynamics in multimode fibers (4 published first-author journal articles; 11 conference presentations).
- Developed and characterized ultrafast single- and multi-mode fiber laser sources, and single-shot ultrafast imaging via spatial multiplexing (685 Gfps).
- Built photonic neural-network and reservoir-computing frameworks (spiking, recurrent, and diffractive) in Python/PyTorch, validated both numerically and experimentally.

PROFEN — Vacuum Electron Device Technologies Department — Istanbul, Turkey
R&D Intern — July 2025 – Sept 2025

- Researched photocathode-based High-Power Microwave (HPM) sources and RF/THz ranging and detection technologies.

Aquila Games — Remote
Game and AI Programmer — March 2020 – Oct 2022

- Engineered NPC AI and core gameplay systems in C# / Unity Engine for *The Sentinel Remake* and *Neon Parkour*.

Journal Articles and Manuscripts

1. **Kesgin, B.U.** & Teğin, U. (2025). Photonic neural networks at the edge of spatiotemporal chaos in multimode fibers. *Nanophotonics* 14(16), 2723–2732. [DOI] (Special Issue: Guided Nonlinear Optics for Information Processing).
2. **Kesgin, B.U.** & Teğin, U. (2024). Implementing the analogous neural network using chaotic strange attractors. *Communications Engineering* 3, 99. [DOI] (Nature Portfolio "Neuromorphic Hardware and Computing" Collection).
3. **Kesgin, B.U.**, Durdu, G.Y. & Teğin, U. (2026). Optical spiking neural networks via rogue-wave statistics. *npj Unconventional Computing* 3, 33. [DOI] (Nature Portfolio "Unconventional Optical Computing" Collection).
4. **Kesgin, B.U.**, Yüce, F. & Teğin, U. (2025). Fiber-based diffractive deep neural network. *Optics Letters* 50, 5254-5257 . [DOI]
5. Eslik, D., **Kesgin, B.U.** et al. (2026). Multimode fiber laser cavities as nonlinear optical processors. Accepted, In press. *Communications Physics*. [arXiv]
6. Eslik, D.*, **Kesgin, B.U.*** & Teğin, U. (2026). Recurrent neural networks implemented through spatiotemporal light propagation in optical fibers. Under review, *Optica*. [arXiv]
7. Eslik, D., **Kesgin, B.U.** & Teğin, U. (2026). Low-cost passive single-shot ultrafast imaging at 685 Gfps. [arXiv]
8. Çarpınlıoğlu, B., **Kesgin, B.U.** & Teğin, U. (2025). Real-time surrogate modeling of nonlinear pulse evolution in multimode fibers. [arXiv]

* Equal contribution.

Patents

Kesgin, B.U. & Teğin, U. (2023). Method and system for enhanced machine learning utilizing chaotic attractors. International Patent (PCT) WO2025063908A1.

Conference Presentations

1. **Kesgin, B.U.**, Yüce, F. & Teğin, U. (2026). Diffractive deep neural networks with multimode fibers. *SPIE Photonics West OPTO*, San Francisco, CA. [DOI]
2. **Kesgin, B.U.** & Teğin, U. (2025). Spatiotemporal chaos-based photonic neural networks. *SPIE Photonics West OPTO*, San Francisco, CA. [DOI]
3. **Kesgin, B.U.** & Teğin, U. (2025). Chaotic photonic reservoir computing with complex dynamics in multimode fibers. *CLEO 2025*, Long Beach, CA. [DOI]
4. **Kesgin, B.U.**, Yüce, F. & Teğin, U. (2025). Diffractive deep neural networks via mechanical mode coupling optimization. *CLEO 2025*, Long Beach, CA. [DOI]
5. **Kesgin, B.U.** & Teğin, U. (2025). Photonic neural networks with multimode fibers at the edge of spatiotemporal chaos. *CLEO Europe 2025*, Munich, Germany. [DOI]
6. **Kesgin, B.U.**, Yüce, F. & Teğin, U. (2025). Diffractive deep neural networks with multimode fibers through mode coupling optimization. *SPIE Optics+Photonics ETAI*, San Diego, CA. [DOI]
7. **Kesgin, B.U.** & Teğin, U. (2025). Chaotic photonic neural networks at the edge of spatiotemporal chaos. *SPIE Optics+Photonics ETAI*, San Diego, CA. [DOI]
8. Çarpınlıoğlu, B., **Kesgin, B.U.** & Teğin, U. (2025). Convolutional neural networks for predicting nonlinear pulse propagation in multimode fibers. *Frontiers in Optics*, Denver, CO. [DOI]
9. Eşlik, D., **Kesgin, B.U.** & Teğin, U. (2026). Passive photonic recurrent neural networks via spatiotemporal multimode fiber dynamics. *SPIE Photonics West OPTO*, San Francisco, CA. [DOI]
10. Eşlik, D., **Kesgin, B.U.** & Teğin, U. (2026). Single-shot imaging achieving 685 billion FPS using spatial multiplexing. *SPIE Photonics West LASE*, San Francisco, CA. [DOI]
11. Çarpınlıoğlu, B., **Kesgin, B.U.** & Teğin, U. (2025). AI-assisted control over spatiotemporal nonlinear fiber dynamics. *SPIE Photonics West OPTO*, San Francisco, CA. [DOI]

Honors, Awards & Fellowships

TÜBİTAK-MAECI Collaborative Project Fellow
(*Project Title: Multiphoton Imaging System with Spatiotemporal Fiber Lasers for Real-Time Optical Biopsy*) 2026 - Present
TÜBİTAK Intern Researcher Program (STAR) Fellow
(*Project Title: Harnessing spatiotemporal nonlinear dynamics of optical cavities for tunable lasers and optical computing*) 2025
Koç University College of Engineering Dean's Honor Roll (×4) 2024–2026
Turkish Ministry of Youth and Sports National Athlete Scholarship 2022–2027

Teaching Experience

Teaching Assistant, EEE 429: Fundamentals of Optics Spring 2025
Teaching Assistant, EEE 206: Electromagnetic Theory Fall 2025

Technical Skills

Experimental: ultrafast fiber laser sources, free-space and fiber optical setups, SLM-based wavefront control, alignment and characterization.
Programming: Python (PyTorch, TensorFlow), C, C++, C#, MATLAB, R.
Tools: LaTeX, Git, Blender, Unity Engine; Arduino, Raspberry Pi.
Languages: Turkish (native), English (IELTS 7.5/9), German (basic), Spanish (basic).

Leadership and Athletic Experience

Turkish Volleyball Federation — National Beach Volleyball Athlete Nov 2019 – Sept 2022
- Competed in 1 World, 3 European, and 4 Balkan Championships (35+ international matches); 17th globally at U19 World Championships.

Koç University Debate Club — Board Member, Captain & Coach July 2023 – June 2024
- Led tournament organization and member training; national tournament finalist.